UGC COMPUTING AWARDS

HANDBOOK

Faculty of Computing

NSBM Green University Town

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Faculty of Computing

The Faculty of Computing provides world class education and training in Computing and Information Technology, both at undergraduate as well as post graduate levels. The Faculty of Computing has also partnered with the world top ranking universities, University College Dublin in Ireland, University of Plymouth in UK and Victoria University in Australia to provide undergraduates with highly recognized International Degrees. The innovative teaching methods along with the latest state of the art equipment form the perfect blend that motivates our students to do their best and reach their goals with ease. The Faculty provides top notch research, training and development services that will help students acquire new knowledge along with the best practices in their respective disciplines. The Faculty of Computing aims to be among the foremost center of excellence in Research and Development (R&D) and advance education in Computing while taking into consideration National as well as Regional requirements for Information and Communication Technology. The Faculty of Computing places equal emphasis on both theory and practice of all aspects of the Computing enabling our students to have sufficient hand on experience to take up any working assignment in their respective IT fields at the end of their degree program. So, look no further for that perfect Computing degree, NSBM's Faculty of Computing is the ideal choice!

Deans Message

Welcome to the Faculty of Computing of the NSBM Green University Town.

NSBM is a dynamic young organization offering innovative educational products to cater to the needs of the fast developing business and industrial economies in the world. Let me congratulate you in becoming part of this dynamic organization.

Your course of study will be up to date and relevant, will be delivered by well qualified staff and it is geared to prepare you for employment. NSBM Graduate profile and student charter aims to develop the students to achieve what they expect in the career path. As students of the NSBM you are expected to work hard and to set high standards. To help you achieve success you are provided with an excellent staff and student support services to help deal with your needs. Our academic, administrative and technical staffs with which you work will be ready to advise and facilitate you. It is your responsibility to take your course of studies very seriously and make the full use of the diverse range of learning opportunities provided to you managing your time effectively in class and in self-directed assignments. Our wish is to see that you become successful in life and be a good ambassador for the university.

Inevitably at the start of all study program you will be shelled with a host of well-intentioned information. Some of that information is important to start your studies immediately and make sure that you are in the right place at the right time. Some information you will need later in your course, whilst other information is about the services the University offers generally which you may need to refer to in the future. We suggest that you download the **NSBM student handbook** and keep it for reference and familiarize yourself with the range of information it contains. This should be the first document of your own e-archive - get into the habit of downloading essential documents like module descriptors and session plans when the course starts.

You are now part of the NSBM family and we look forward to working with you to help you to succeed as an NSBM Graduate.

Very best wishes,

Dean

Faculty of Computing

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University Grants Commission & Ministry of Higher education approved degrees offered by NSBM Green University Town

BSc in Management Information Systems (Special)

BSc (Hons) in Software Engineering

BSc (Hons) in Computer Science

BSc (Hons) in Computer Networks

BSc in Multimedia

Bachelor of Science in Management Information Systems (Special)

BSc in Management Information Systems (Special) program at NSBM Faculty of Computing is a carefully designed interdisciplinary program focusing on both computing and business aspect. The MIS graduate aims to leverage technology, people and processes to improve an organization's operational efficiency and to make substantive contributions to the use of information systems in corporate decision making.

Overview of the Program

This four-year program specifically targets to produce IS professionals with the required knowledge and skills. During the first academic year, students are learning the fundamentals of computing which is essential to advance their knowledge in any computing discipline. During the second year of studies, students are learning demanding topics in the computing industry. During the third year, students learn IT-business oriented modules along with mandatory industry placement. During their final year, students are provided with knowledge and understanding of diverse subject domains and an individual award project that allow them to showcase their learning over the years.

Key Features:

UGC Approved 4-year honours degree amalgamating computing and business disciplines

A diverse choice of elective modules to choose from, in their final year of studies, to cater to individual preferences and ambitions

Includes mandatory industry placement and supervised research projects element

Focuses on infusing entrepreneurial skills

AWARD

BSc in Management Information Systems (Special)

DURATION

4 Years

ACADEMIC PROGRESSION

MSc, MBA, PhD

CAREER PROSPECTS

Software Engineer, Business Analyst, Quality Assurance Engineer, UI Engineer, System Support Engineer, Web Design, IT Management Interns, Project managers

Program Content: */**

YEAR 01

Module Code	Module Name	Credit Value	Type
CS101.3	Introduction to Computer Science	3	Core
MA101.3	Mathematics for Computing	3	Core
CS102.3	Programming in C	3	Core
CS103.3	Professional Development	3	Core
CN101.3	Data Communication and Networks	3	Core
SE101.3	Object Oriented Programming with Java	3	Core
CS106.3	Algorithms and Data Structures	3	Core
CS104.3	Computer Architecture	3	Core
CS105.3	Database Management Systems	3	Core
SE102.3	Web Based Application Development	3	Core

YEAR 02

Module Code	Module Name	Credit Value	Type
MA201.3	Statistics for Computing	3	Core
SE201.3	Systems Analysis and Design	3	Core
SE204.3	Development of Enterprise Applications I	3	Core
IS202.3	Foundations of Information Systems	3	Core
BS201.3	Business Studies	3	Core
SE202.3	Introduction to Software Engineering	3	Core
CN201.3	Computer Networks	3	Core
SE206.3	Human Computer Interaction	3	Core
IS204.3	Business Process Management	3	Core
IS202.3	Enterprise Systems	3	Core

YEAR 03

Module Code	Module Name	Credit Value	Type
CS301.3	IT Project Management	3	Core
CS304.3	Advanced Database Management Systems	3	Core
SE303.3	Mobile Application Development	3	Elective
SE305.6	Internship[#]	6	Core
SE307.3	Social Issues and Professional Practice	3	Core
IS302.3	IT Audit and Control	3	Elective
IS301.3	Enterprise Architecture	3	Core
CN302.3	Wireless Technologies and Network Programming	3	Elective
CS306.3	Information Assurance and Security	3	Elective
SE304.3	Software Quality Assurance	3	Elective
SE308.3	Software Process Management	3	Elective

YEAR 04

Module Code	Module Name	Credit Value	Type
BS401.3	Business Policy and Strategy	3	Elective
BS402.3	Entrepreneurship	3	Elective
CN402.3	Enterprise Networks	3	Core
IS401.3	Management Information Systems	3	Core
IS405.6	MIS Special Award Project	6	Core
SE402.3	Development of Enterprise Applications II	3	Elective
IS404.3	Business Analytics	3	Elective
CN406.3	Disaster Recovery and High Availability Techniques	3	Elective
CS405.3	Data Warehousing and Data Mining	3	Elective
CS407.3	Internet of Things	3	Elective
SE403.3	Platform Based Development	3	Elective
SE404.3	Agent Based Systems	3	Elective
IS402.3	E-Business Application Development	3	Elective

[#] *Internship module does not contribute to the GPA calculation for the determination of the Award classification but contributes to the credit requirement at Level 3.*

***** *Course content may update periodically*

****** *Elective modules will be offered with the prior approval of the faculty board*

Bachelor of Science (Honours) in Software Engineering

BSc (Honours) in Software Engineering program at NSBM Faculty of Computing is carefully designed in line with the latest ACM and IEEE guidelines for Software Engineering undergraduate awards and aims to produce competent and well molded computer professionals who can engineer the design and construction of complex software systems.

OVERVIEW OF THE PROGRAM

The program is structured in a way to provide students with a broader computing perspective in the first year and focusing on the core software engineering body of knowledge in the second and third years. Students also undergo compulsory industry training in the third year. In the fourth year, students mould with knowledge and understanding of diverse subject domains and an individual award project that allow them to showcase their learning over the years.

KEY FEATURES

UGC Approved 4-year honours degree

A diverse choice of elective modules to choose from, in their final year of studies, to cater to individual preferences and ambitions

Includes mandatory industry placement and supervised research projects element

Focuses on infusing knowledge and skills to develop and maintain Computer Networks within an organization or between organizations

AWARD

BSc (Honours) in Software Engineering

DURATION

4 Years

ACADEMIC PROGRESSION:

MSc, MBA, PhD

CAREER PROSPECTS:

Software Engineer, Systems Analyst, Quality Assurance Engineer, UI Engineer, System Support Engineer, Web Design

Program Content: */**

YEAR 01

Module Code	Module Name	Credit Value	Type
CS101.3	Introduction to Computer Science	3	Core
MA101.3	Mathematics for Computing	3	Core
CS102.3	Programming in C	3	Core
CS103.3	Professional Development	3	Core
CN101.3	Data Communication and Networks	3	Core
SE101.3	Object Oriented Programming with Java	3	Core
CS106.3	Algorithms and Data Structures	3	Core
CS104.3	Computer Architecture	3	Core
CS105.3	Database Management Systems	3	Core
SE102.3	Web Based Application Development	3	Core

YEAR 02

Module Code	Module Name	Credit Value	Type
CS201.3	Operating Systems	3	Core
SE202.3	Introduction to Software Engineering	3	Core
MA201.3	Statistics for Computing	3	Core
SE201.3	Systems Analysis and Design	3	Core
CS203.3	Algorithms and Complexity	3	Core
CN201.3	Computer Networks	3	Core
SE204.3	Development of Enterprise Applications I	3	Core
SE205.3	Software Architecture	3	Core
SE206.3	Human Computer Interaction	3	Core
IS201.3	Business Processes and ERP	3	Core

YEAR 03

Module Code	Module Name	Credit Value	Type
MA301.3	Advanced Mathematics for Computing	3	Core
SE304.3	Software Quality Assurance	3	Core
SE307.3	Social Issues and Professional Practice	3	Core
SE308.3	Software Process Management	3	Core
CS306.3	Information Assurance and Security	3	Core
SE309.3	Software Verification and Validation	3	Core
SE305.6	Internship#	6	Core
CS304.3	Advanced Database Management Systems	3	Elective
SE303.3	Mobile Application Development	3	Elective
CS301.3	IT Project Management	3	Elective

YEAR 04

Module Code	Module Name	Credit Value	Type
BS401.3	Business Policy and Strategy	3	Core
CN402.3	Enterprise Networks	3	Elective
IS401.3	Management Information Systems	3	Elective
SE402.3	Development of Enterprise Applications II	3	Elective
SE401.6	SE Honours Award Project	6	Core
CS403.3	Artificial Intelligence	3	Elective
CS408.3	Embedded Systems	3	Elective
CS402.3	Computer Graphics and Visualization	3	Elective
CS404.3	Parallel and Distributed Computing	3	Elective
CS405.3	Data Warehousing and Data Mining	3	Elective
CS407.3	Internet of Things	3	Elective
SE403.3	Platform Based Development	3	Elective
SE404.3	Agent Based Systems	3	Elective
IS402.3	E-Business Application Development	3	Elective

[#] *Internship module does not contribute to the GPA calculation for the determination of the Award classification but contributes to the credit requirement at Level 3.*

***** *Course content may update periodically*

****** *Elective modules will be offered with the prior approval of the faculty board*

Bachelor of Science (Honours) in Computer Science

BSc (Honours) in Computer Science Program at NSBM Faculty of Computing is carefully designed in line with the latest ACM and IEEE guidelines for Computer Science undergraduate awards and aims to produce competent and well moulded computer professionals who have sound technical foundation in computer science and the ability to creatively apply computer and related technologies to practical problems.

Overview of the Program

The Program is structured in a way to provide students with a broader computing perspective in the first year and focusing on the core Computer Science body of knowledge in the second and third years. Students also undergo compulsory industry training in the third year. In the fourth year, students are provided with knowledge and understanding of diverse subject domains and an individual award project that allow them to showcase their learning over the years.

Key Features:

UGC Approved 4-year honours degree

A diverse choice of elective modules to choose from, in their final year of studies, to cater to individual preferences and ambitions

Includes mandatory industry placement and supervised research projects element

Focuses on developing knowledge and skills in breadth and depth

AWARD

BSc (Honours) in Computer Science

DURATION

4 Years

ACADEMIC PROGRESSION:

MSc, MBA, PhD

CAREER PROSPECTS:

Analysts, Designers, Architects, Developers, Testing and QA Specialists, System Administrators, Business/technology Analysts and System Integrators.

Program Content: */**

YEAR 01

Module Code	Module Name	Credit Value	Type
CS101.3	Introduction to Computer Science	3	Core
MA101.3	Mathematics for Computing	3	Core
CS102.3	Programming in C	3	Core
CS103.3	Professional Development	3	Core
CN101.3	Data Communication and Networks	3	Core
SE101.3	Object Oriented Programming with Java	3	Core
CS106.3	Algorithms and Data Structures	3	Core
CS104.3	Computer Architecture	3	Core
CS105.3	Database Management Systems	3	Core
SE102.3	Web Based Application Development	3	Core

YEAR 02

Module Code	Module Name	Credit Value	Type
MA201.3	Statistics for Computing	3	Core
SE201.3	Systems Analysis and Design	3	Core
SE204.3	Development of Enterprise Applications I	3	Core
SE205.3	Software Architecture	3	Core
CS202.3	Systems Fundamentals	3	Core
SE202.3	Introduction to Software Engineering	3	Core
CN201.3	Computer Networks	3	Core
SE206.3	Human Computer Interaction	3	Core
CS203.3	Algorithms and Complexity	3	Core
CS201.3	Operating Systems	3	Core

YEAR 03

Module Code	Module Name	Credit Value	Type
CS304.3	Advanced Database Management Systems	3	Core
SE303.3	Mobile Application Development	3	Elective
SE305.6	Internship[#]	6	Core
SE307.3	Social Issues and Professional Practice	3	Core
MA301.3	Advanced Mathematics for Computing	3	Core
CS303.3	Computational Theory	3	Core
CS305.3	Programming Languages and Compiler Design	3	Core
CN302.3	Wireless Technologies and Network Programming	3	Elective
CS306.3	Information Assurance and Security	3	Core
SE308.3	Software Process Management	3	Elective
CS302.3	Cryptography	3	Elective

YEAR 04

Module Code	Module Name	Credit Value	Type
BS401.3	Business Policy and Strategy	3	Elective
BS402.3	Entrepreneurship	3	Elective
SE402.3	Development of Enterprise Applications II	3	Elective
CS403.3	Artificial Intelligence	3	Core
CS408.3	Embedded Systems	3	Elective
CS402.3	Computer Graphics and Visualization	3	Core
CS404.3	Parallel and Distributed Computing	3	Core
CS405.3	Data Warehousing and Data Mining	3	Elective
CS407.3	Internet of Things	3	Elective
SE403.3	Platform Based Development	3	Elective
SE404.3	Agent Based Systems	3	Elective
CS401.6	CS Honours Award Project	6	Core
CS406.3	Bio Informatics	3	Elective

[#] *Internship module does not contribute to the GPA calculation for the determination of the Award classification but contributes to the credit requirement at Level 3.*

***** *Course content may update periodically*

****** *Elective modules will be offered with the prior approval of the faculty board*

Bachelor of Science (Honours) in Computer Networks

BSc (Honours) in Computer Networks program at NSBM Faculty of Computing is carefully designed in line with the latest subject benchmarks for Computer Networks undergraduate awards and aims to produce competent and well moulded network professionals who are capable of developing and managing complex network systems.

Overview of the Program

This is a four-year degree program that aims to produce graduates with required skills and knowledge in the core and emerging subject areas in Computer Networks. Students also undergo compulsory industry training in the third year and the fourth year, students are provided with knowledge and understanding of diverse subject domains and an individual award project that allow them to showcase their learning over the years.

Key Features:

UGC Approved 4-year honours degree

A diverse choice of elective modules to choose from, in their final year of studies, to cater to individual preferences and ambitions

Includes mandatory industry placement and supervised research projects element

Focuses on infusing knowledge and skills to develop and maintain Computer Networks within an organization or between organizations

AWARD

BSc (Honours) in Computer Networks

DURATION

4 Years

ACADEMIC PROGRESSION

MSc, MBA, PhD

CAREER PROSPECTS

Network Administrator, Network Engineer, Network Analyst, IT/Systems Support Engineer, Network Architect, VoIP Engineer.

Program Content: */**

YEAR 01

Module Code	Module Name	Credit Value	Type
CS101.3	Introduction to Computer Science	3	Core
MA101.3	Mathematics for Computing	3	Core
CS102.3	Programming in C	3	Core
CS103.3	Professional Development	3	Core
CN101.3	Data Communication and Networks	3	Core
SE101.3	Object Oriented Programming with Java	3	Core
CS106.3	Algorithms and Data Structures	3	Core
CS104.3	Computer Architecture	3	Core
CS105.3	Database Management Systems	3	Core
SE102.3	Web Based Application Development	3	Core

YEAR 02

Module Code	Module Name	Credit Value	Type
MA201.3	Statistics for Computing	3	Core
SE201.3	Systems Analysis and Design	3	Core
SE204.3	Development of Enterprise Applications I	3	Core
CN202.3	Network Management	3	Core
CN205.3	Voice and Video over IP	3	Core
CN201.3	Computer Networks	3	Core
SE206.3	Human Computer Interaction	3	Core
CS201.3	Operating Systems	3	Core
CN203.3	Network Systems Configuration and Administration	3	Core
CN204.3	Network Security	3	Core

YEAR 03

Module Code	Module Name	Credit Value	Type
CS301.3	IT Project Management	3	Core
SE305.6	Internship[#]	6	Core
IS301.3	Enterprise Architecture	3	Elective
MA301.3	Advanced Mathematics for Computing	3	Elective
CN301.3	Advanced Routing and Switching	3	Core
CN303.3	Service Provider Systems and Networks	3	Core
CN302.3	Wireless Technologies and Network Programming	3	Core
CS306.3	Information Assurance and Security	3	Core
CS302.3	Cryptography	3	Elective
IS302.3	IT Audit and Control	3	Elective
SE307.3	Social Issues and Professional Practice	3	Core

YEAR 04

Module Code	Module Name	Credit Value	Type
BS401.3	Business Policy and Strategy	3	Elective
BS402.3	Entrepreneurship	3	Elective
CN402.3	Enterprise Networks	3	Core
IS401.3	Management Information Systems	3	Elective
CN406.3	Disaster Recovery and High Availability Techniques	3	Core
CS408.3	Embedded Systems	3	Elective
CS404.3	Parallel and Distributed Computing	3	Elective
CS405.3	Data Warehousing and Data Mining	3	Elective
CS407.3	Internet of Things	3	Elective
SE403.3	Platform Based Development	3	Elective
IS402.3	E-Business Application Development	3	Elective
CN401.6	CN Honours Award Project	6	Core
CN403.3	Intrusion Prevention, Detection & Response	3	Core

[#] *Internship module does not contribute to the GPA calculation for the determination of the Award classification but contributes to the credit requirement at Level 3.*

***** *Course content may update periodically*

****** *Elective modules will be offered with the prior approval of the faculty board*

Bachelor of Science in Multimedia

BSc in Multimedia is a three-year degree program offered by NSBM with the approval of the UGC. This program focuses on idea-based education and main emphasis is given for conceptualization and ideation through the design process.

Overview of the Program

This is a multi-disciplinary course providing the students with a comprehensive education on the design process, research, management, production and finally evaluation. The first year will give them opportunity to explore their creativity, thinking patterns while learning the key concepts of design. The second year will be a combination of lectures and different types of practical sessions where they will learn the practical applications of knowledge. Third year has a compulsory industry training to apply the learnt theories, principles and practical knowledge in real life scenarios. Further they get to learn about the diverse subject domains and conduct an individual project from initiation to execution.

Key Features

UGC Approved 3-year degree

A diverse area of learning with theoretical knowledge and practical experience

Many different job opportunities in the Multimedia sector with a lot of scope for new developments and a promise for a highly acclaimed career

Practice the learnt design knowledge through the mandatory 90-day internship program in the 3rd year

AWARD

BSc in Multimedia

DURATION

3 Years

ACADEMIC PROGRESSION:

MSc, PhD

CAREER PROSPECTS:

Graphic and Communication Designer, Art Director, Creative Director, Video Editor, Production Director, 2D and 3D Visualizer, Animation Designer, Virtual Environmental Designer, Game Designer, Film Director, Web and interface designer, Entrepreneur

Program Content: */**

YEAR 01

Module Code	Module Name	Credit Value	Type
MM 1001	Concept Development	3	Core
MM 1002	World of Multimedia	2	Core
MM 1003	Computing Fundamentals	2	Core
MM 1004	Drawing	2	Core
MM 1005	Design Principles 1	3	Core
MM 1006	Typography Design	3	Core
MM 1007	Illustration	2	Core
MM 1008	Story Telling and Script Writing	2	Core
MM 1009	Art, Design and society	2	Core
MM 1010	Animation Studies	3	Core
MM 1011	Digital Imaging	3	Core
MM 1012	Multimedia Imaging	3	Core

YEAR 02

Module Code	Module Name	Credit Value	Type
MM 1013	Photography	2	Core
MM 1014	Cinematography	3	Core
MM 1015	3D Design 1	2	Core
MM 1016	Game Design	2	Core
MM 1017	Video Editing and Production	3	Core
MM 1018	Interactive Design	3	Core
MM 1019	Advertising and Media Art	3	Core
MM 1020	Multimedia Project Management	3	Core
MM 1021	3D Design 2	2	Core
MM 1022	Web Design	2	Core
MM 1023	Design Principles 2	2	Core
MM 1024	Visual Communication	3	Core

YEAR 03

Module Code	Module Name	Credit Value	Type
MM 1025	Multimedia Production	2	Core
MM 1026	Professional Studies	3	Core
MM 1027	Visual Effects Production	3	Core
MM 1028	Research Methodology	2	Core
MM 1029	Dissertation	5	Core
MM 1030	Internship[#]	4	Core
MM 1031	Media Law and Ethics	2	Core
MM 1032	Entrepreneurship	3	Core
MM 1033	Final Project	6	Core